

Progressing Wave Energy Conversion via Large-Scale Additive Manufacturing (WEC-pro)

Work Package 2 Report:
Structural toolbox for LSAM incorporation



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Contents

Executive summary	5
1 Introduction and Overview	6
1.1 Project Context and WP2 Objectives	6
1.2 Case Study Update: Transition to the Point Absorber (PA)	6
1.3 Scope of the Present Report	7
2 Summary of WP1 Hydrodynamic Optimisation for the PA	7
2.1 Monte Carlo Framework and LSAM Filtering	7
2.2 Hydrodynamic Evaluation and Optimal Selection	8
3 The LFAM Structural Toolbox	11
3.1 Software Selection: FreeCAD vs. Salome-Meca	11
3.2 Defining the “Toolbox” Concept	12
4 Structural Design of the Point Absorber for LFAM	15
4.1 Hydrodynamic Design and Geometry Initialization	15
4.2 Internal Structural Engineering and Ballast Chambers	15
4.3 Design for Additive Manufacturing (DfAM) Constraints	17
5 Preliminary Finite Element Method (FEM) Analysis	17
5.1 Material Specifications: PP GF30 (FGF)	18
5.2 Definition of Load Cases	19
5.3 Preliminary Results and Verification	20
5.4 PA and Modelling Specifications	27

6 Conclusion and Integration with WP3

27

List of Figures

1	Automated Monte Carlo simulation framework developed in WP1, highlighting the integration of LSAM validation filters prior to hydrodynamic evaluation.	8
2	Visualisation of the 1,130 accepted B-spline profiles that successfully passed the 60° overhang and monotonicity LSAM filters.	9
3	The optimal geometry (spline 282) selected for its maximum AEP/wet surface ratio, tailored to the Reggio Calabria wave climate.	10
4	Hydrodynamic properties of spline 282: added mass, added damping, and wave excitation force coefficients across the frequency domain. . .	10
5	Heave Response Amplitude Operator (RAO) of the optimal PA floater (spline 282), indicating highest resonance at about $\omega = 5$ rad/sec. . .	11
6	Sequential workflow of the conceptual LFAM Structural Toolbox protocol.	15
7	Cross-sectional view of the FreeCAD structural model, detailing the outer hull, the central heave-constraining slot, internal ballast chambers, and top deck sensor access ports.	16
8	External view of the Von Mises stress distribution under 30 kPa hydrostatic pressure (LC1).	21
9	Internal view of the Von Mises stress distribution under 30 kPa hydrostatic pressure (LC1).	21
10	Side view of the Von Mises stress distribution under a 15 kN lateral load (LC2).	22
11	Internal view of the Von Mises stress distribution under a 15 kN lateral load (LC2).	22
12	Internal view of the Von Mises stress distribution under a 15 kN lateral load applied at 45 degrees (LC3).	23
13	Bottom view of the Von Mises stress distribution under a 21 kN bottom load (LC4).	24
14	Lateral view of the Von Mises stress distribution under a 21 kN bottom load (LC4).	24
15	Internal view of the Von Mises stress distribution under a 21 kN bottom load (LC4).	25

16 Bottom view of the Von Mises stress distribution under a 21 kN top
load (LC5). 26

17 Internal view of the Von Mises stress distribution under a 21 kN top
load (LC5). 26

List of Tables

1 Material Configuration Table: PP GF30 (FGF) 19

2 Definition of Operational Load Cases for FEM Verification 20

3 Full Specifications of the Optimised Point Absorber (PA) 27

Executive summary

This document details the activities and outcomes of Work Package 2 (WP2) of the WEC-pro project. Building upon the hydrodynamic optimisation achieved in WP1, WP2 focuses on the structural design and analysis required to take advantages of Large-Format Additive Manufacturing (LFAM) into the development cycle of Wave Energy Converters (WECs).

A key strategic update in this phase was the substitution of the PivWEC case study with a Point Absorber (PA) geometry, leveraging the extensive real-sea deployment experience of the MORE Lab at POLITO. This report establishes the "LFAM Structural Toolbox," formulated not as a standalone custom software, but as a digital protocol utilizing the FreeCAD platform. It details the transition from the optimised outer geometry - for efficient hydrodynamics - to a fully printable efficient structural design. Key developments include the integration of internal ballast chambers respecting strict LFAM overhang limits, sensor access ports, a central heave-constraining slot, and the definition of the minimum structural thickness and the upside-down printing orientation. Finally, the report outlines the preliminary Finite Element Method (FEM) verification using PP GF30 (FGF) material properties across five distinct operational load cases. The experimental validation of these physical models is subsequently addressed in WP3.

1 Introduction and Overview

1.1 Project Context and WP2 Objectives

The global push towards sustainable energy has highlighted the immense potential of wave energy. However, the commercialisation of Wave Energy Converters (WECs) remains bottlenecked by high Research and Development (R&D) costs, lengthy prototyping phases, and the reliance on traditional, rigid manufacturing methods (e.g., steel welding, composite moulding). The WEC-pro project proposes a paradigm shift by integrating LFAM into the WEC development cycle, enabling rapid, tool-less, and cost-effective iterations.

While Work Package 1 (WP1) successfully developed numerical tools to optimise the hydrodynamic shapes of WECs for maximum power absorption, these idealised geometries must be structurally viable for both the harsh marine environment and the specific constraints of the 3D printing process. Therefore, the primary objective of Work Package 2 (WP2) is to adjust and apply structural design and analysis methods to improve WEC structural efficiency specifically for LFAM application.

The main objectives relative to WP2 involves:

- Defining strict LFAM design constraints (e.g., minimum printable thicknesses, overhang angles, and support structure minimisation).
- Developing a "Structural Toolbox" to bridge the gap between hydrodynamic profiles and printable CAD models.
- Performing preliminary Finite Element Method (FEM) analysis to ensure structural reliability, material efficiency, and load-bearing capacity before physical manufacturing.

1.2 Case Study Update: Transition to the Point Absorber (PA)

In the initial plan, the rotating PivWEC was slated alongside the PeWEC as one of the two primary case studies. However, during the execution of WP1 and the transition into WP2, a strategic decision was made to substitute the PivWEC with a Point Absorber (PA) WEC configuration.

This substitution was driven by the maturity of the technology and the specific expertise available at the host institution. The PivWEC is currently at a very early stage of development (TRL 3), which introduces excessive uncertainties regarding its

internal mechanical layout and Power Take-Off (PTO) constraints. Conversely, the MORE Lab at Politecnico di Torino (PoliTo) possesses extensive, hands-on knowledge of Point Absorber technologies. This expertise is anchored by the successful real-scale deployment of the WEPA (Wave Energy Point Absorber) device at sea in Alghero, Italy. By aligning the WP2 structural design efforts with the PA archetype, the project benefits from validated real-world operational data, mature PTO integration concepts, and a higher baseline TRL. Such decision was made especially in the view of potential future work lines.

1.3 Scope of the Present Report

This report documents the translation of the hydrodynamically optimised PA floater, developed via stochastic Monte Carlo simulations in WP1 to maximise the Annual Energy Production (AEP) to wet-surface ratio—into a fully functional, LFAM-ready structural model.

The core of this WP is the establishment of the **LFAM Structural Toolbox**. Rather than developing a bespoke software application, this "toolbox" is defined as a highly structured, open-source-driven *protocol* for CAD modelling and FEM analysis using the FreeCAD platform. The report details the internal engineering required for physical testing, such as internal ballast chambers tailored to LFAM overhang limits, integrated sensor access pathways, and a central vertical slot to constrain the physical model to heave-only motion. Finally, the report outlines the preliminary structural verification across five distinct operational load cases using the specific material properties of PP GF30.

2 Summary of WP1 Hydrodynamic Optimisation for the PA

Before detailing the structural design in WP2, it is essential to summarise the hydrodynamic optimisation carried out in WP1, as this defined the outer envelope of the Point Absorber. Unlike evolutionary algorithms used for other WEC types, the PA development employed a fully automated Monte Carlo simulation framework.

2.1 Monte Carlo Framework and LSAM Filtering

The computational workflow was designed to handle large datasets of potential designs while strictly enforcing manufacturability from the earliest stages. As il-

illustrated in Figure 1, the process initiated with the stochastic generation of 50,000 random hull profiles defined by cubic B-splines.

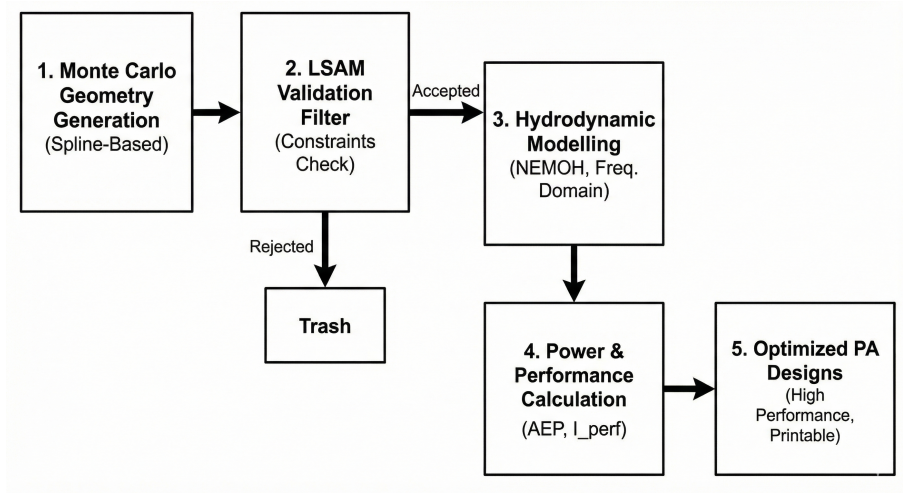


Figure 1: Automated Monte Carlo simulation framework developed in WP1, highlighting the integration of LSAM validation filters prior to hydrodynamic evaluation.

These 50,000 candidates were immediately subjected to a rigorous LSAM validation filter. Geometries were discarded if they violated printability constraints, specifically:

- **Monotonicity:** The curve must be continuous without self-intersections or re-entrant pockets in the Z and X axes, preventing shapes that would trap water or be impossible to extrude.
- **Overhang Limits:** To ensure the shape could be 3D printed without collapsing, a maximum overhang angle of 60° was strictly enforced.

2.2 Hydrodynamic Evaluation and Optimal Selection

Out of the initial 50,000 generated splines, only 1,130 shapes successfully passed the LSAM printability constraints (Figure 2). These accepted geometries were then passed to the NEMOH Boundary Element Method (BEM) solver to compute added mass, radiation damping, and excitation forces.

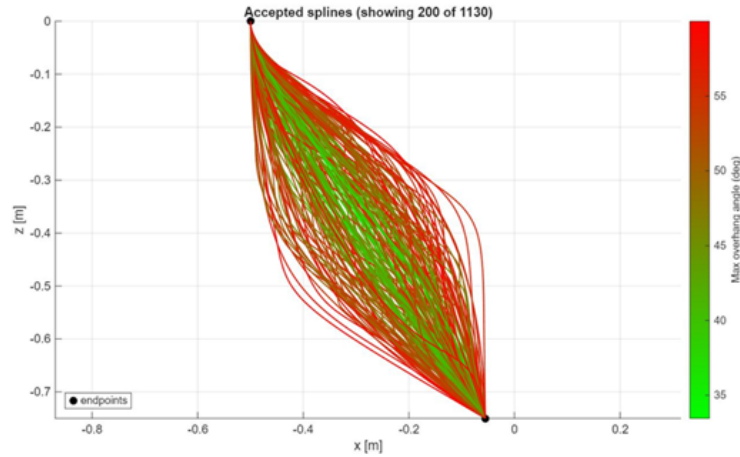


Figure 2: Visualisation of the 1,130 accepted B-spline profiles that successfully passed the 60° overhang and monotonicity LSAM filters.

The energy performance was evaluated by integrating power output across the specific wave climate of the NOEL laboratory deployment site (Reggio Calabria, Italy) to determine the AEP. To maximise efficiency relative to material usage (a critical factor for LFAM cost-effectiveness), the final optimal geometry was selected based on an Index of Performance (I_{perf}):

$$I_{perf} = \frac{AEP}{\text{Wet Surface Area}} \quad [\text{MWh/m}^2] \quad (1)$$

Of all the evaluated splines, **spline 282** emerged as the most performing geometry, achieving the highest AEP to wet-surface area ratio tailored specifically for the wave climate of Reggio Calabria. Consequently, this specific spline shape was selected as the definitive baseline for the structural development in WP2.

Figure 3 illustrates the isolated, optimal profile of spline 282. The fundamental hydrodynamic properties of this selected geometry—specifically the frequency-dependent added mass, radiation damping, and wave excitation force coefficients computed via NEMOH—are presented in Figure 4. Furthermore, the dynamic response of the floater is characterised by its Heave Response Amplitude Operator (RAO), shown in Figure 5, which demonstrates the floater’s excellent tuning to the dominant wave frequencies of the deployment site.

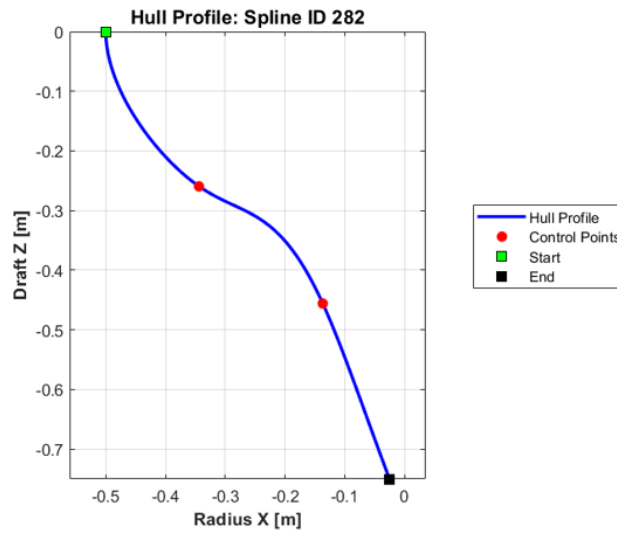


Figure 3: The optimal geometry (spline 282) selected for its maximum AEP/wet surface ratio, tailored to the Reggio Calabria wave climate.

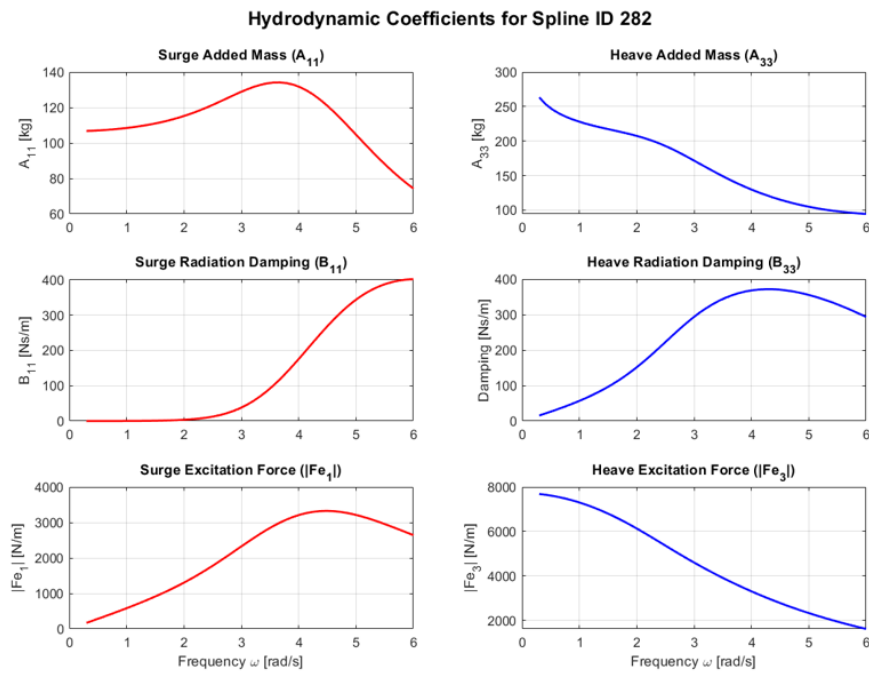


Figure 4: Hydrodynamic properties of spline 282: added mass, added damping, and wave excitation force coefficients across the frequency domain.

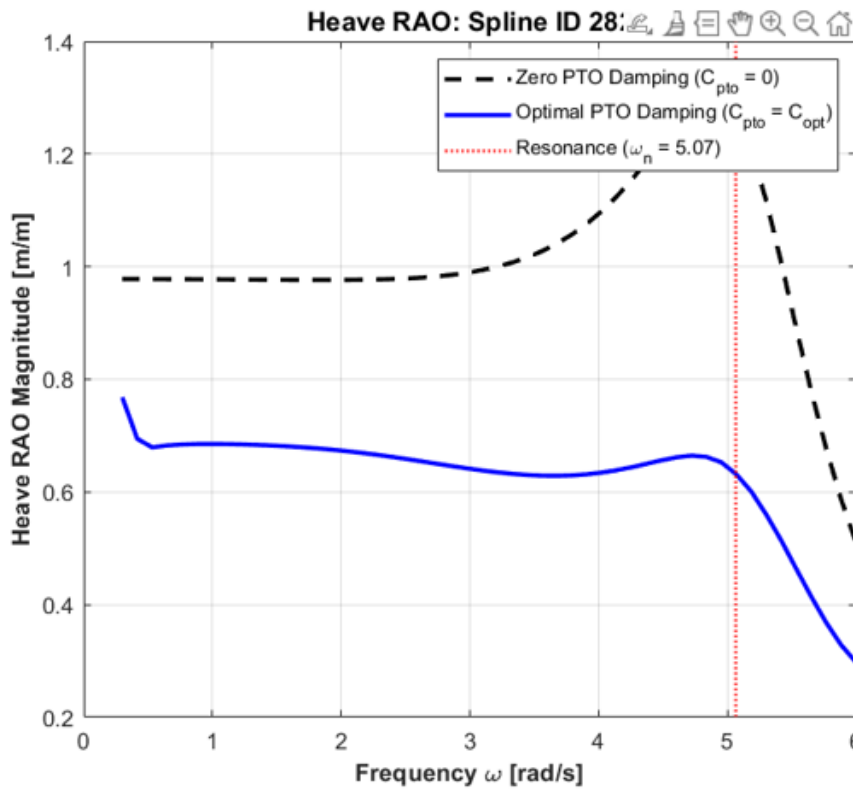


Figure 5: Heave Response Amplitude Operator (RAO) of the optimal PA floater (spline 282), indicating highest resonance at about $\omega = 5$ rad/sec.

3 The LFAM Structural Toolbox

3.1 Software Selection: FreeCAD vs. Salome-Meca

To adhere to the open-source mandate of the WEC-pro project, two primary platforms were evaluated for hosting this structural protocol: FreeCAD and Salome-Meca (incorporating the Code_Aster solver).

Salome-Meca (Code_Aster)

- *Pros:* It is a highly powerful open-source suite widely used in heavy civil and nuclear engineering. It possesses advanced capabilities, most notably the ability to perform complex anisotropic FEM modelling, which is highly relevant for accurately capturing the layer-by-layer mechanical realities of LFAM components. It also handles complex multi-physics simulations exceptionally well.
- *Cons:* The software features a highly complex and non-intuitive user interface. Furthermore, much of the advanced technical documentation and active com-

munity support is exclusively in French. The steep learning curve proved too significant to maintain the agile prototyping timeline required by the project.

FreeCAD

- *Pros:* FreeCAD offers a fully integrated, Python-driven parametric environment. It features dedicated, easy-to-use workbenches for every step of the process: the *Sketcher* and *Part Design* modules for robust geometry creation, the *Draft* module for 2D to 3D conversions, and the *Mesh* and *FEM* modules (utilizing the CalculiX solver) for structural verification. Additionally, its macro capabilities allow for the automation of repetitive tasks, and it boasts a massive, global, English-speaking support community.
- *Cons:* The native FEM workbench primarily handles isotropic materials, meaning the anisotropic nature of 3D printing must be approximated using bounding configurations. Additionally, it can occasionally suffer from topological naming issues during complex boolean operations, though this is actively being improved by the community.

Given the practical implementation barriers of Salome-Meca, **FreeCAD** was selected as the foundational platform for the toolbox. It provides a robust, user-friendly, and highly reproducible environment perfectly suited for the preliminary structural sizing and LFAM geometric preparation required in WP2.

3.2 Defining the “Toolbox” Concept

In the context of the WEC-pro project, the “Structural Toolbox” was conceptualised not as a rigid, standalone executable code, but rather as an adaptable, digital engineering protocol. Rather than acting as a “black-box” software, this conceptual toolbox is a transparent, step-by-step methodology that guides the engineer through the necessary Computer-Aided Design (CAD) and Computer-Aided Engineering (CAE) phases required to prepare a WEC floater for 3D printing.

This protocol establishes a standardized workflow connecting the purely mathematical hydrodynamic surfaces generated in WP1 with the physical manufacturing constraints of LFAM. By formalizing this process, the toolbox ensures that the transition from a hydrodynamic envelope to a printable structure is reproducible, structurally sound, and compliant with open-source engineering principles.

The conceptual toolbox follows a sequential nine-step pipeline designed to iteratively refine both the structural integrity and the physical printability of the WEC device. The detailed breakdown of each step is as follows:

- **Step 1: Importation and 3D Revolution.** The purely mathematical 2D B-spline profiles—representing the optimal hydrodynamic envelope generated in WP1—are imported into the CAD environment. These profiles are then revolved around the central axis to create the foundational 3D solid bodies, establishing the baseline volumetric footprint of the floater.
- **Step 2: Topology Optimization.** To understand the most efficient structural design, topology optimization is performed under critical hydrodynamic load cases. This step identifies the primary stress vectors and load paths, highlighting regions where material is strictly necessary for structural survival and areas where mass can be safely removed to save printing time and costs.
- **Step 3: Application of DfAM Principles.** Guided by the topology optimization results, Design for Additive Manufacturing (DfAM) principles are actively applied to shape the internal and external geometry. This involves a comprehensive suite of parametric CAD operations—including, but not limited to, 2D sketching, drafting, lofting, extruding, and complex boolean operations (cuts, intersections, and unions)—to translate the theoretical load paths into printable physical features.
- **Step 4: Minimal Thickness Definition.** The minimal structural wall thickness for all CAD components is explicitly defined. This dimensioning incorporates appropriate safety factors to withstand hydrostatic pressure and wave slamming, while simultaneously accounting for the physical limitations of the LFAM extrusion nozzle (e.g., ensuring the wall thickness is a clean multiple of the printed bead width).
- **Step 5: Integrated Ballast Chamber Design.** Internal ballast chambers are engineered to serve a dual purpose. Primarily, they hold the necessary mass to achieve the required hydrodynamic draft. Structurally, they act as internal stiffening ribs that reinforce the outer hull. Crucially, these chambers are specifically tailored to comply with AM overhang constraints (e.g., maintaining self-supporting angles of $45^\circ - 60^\circ$) to eliminate the need for internal sacrificial supports.
- **Step 6: Structural Meshing.** The resulting complex, multi-featured geometry is discretized into a high-quality structural mesh. This step ensures proper node connectivity across all internal junctions (such as where the ballast walls meet the outer hull) in preparation for numerical analysis.

- **Step 7: FEM Verification and Feedback Loop.** A rigorous Finite Element Method (FEM) analysis is conducted to verify that the structure can withstand both standard operational forces and extreme wave loads. If the extracted stress concentrations exceed the material's limits (particularly the weaker Z-axis inter-layer adhesion limit), an iterative feedback loop is triggered, prompting a return to Step 3 for targeted design revisions.
- **Step 8: Final CAD Resolution.** Once the structure passes FEM verification, the CAD model undergoes a final cleanup. This involves resolving any topological naming issues (a common requirement in parametric modelling), merging overlapping features, and ensuring that all combined solid bodies form a perfectly watertight (manifold) volume without any geometric singularities.
- **Step 9: LFAM Surface Conversion.** In the final step, the verified and watertight 3D solid bodies are tessellated and converted into surface mesh formats (e.g., high-resolution STL or STEP files). These files act as the definitive digital handover, fully compatible with the LFAM robotic slicing software used to generate the physical extrusion toolpaths.

The complete workflow is illustrated in the coded scheme below (Figure 6), encompassing these key stages:

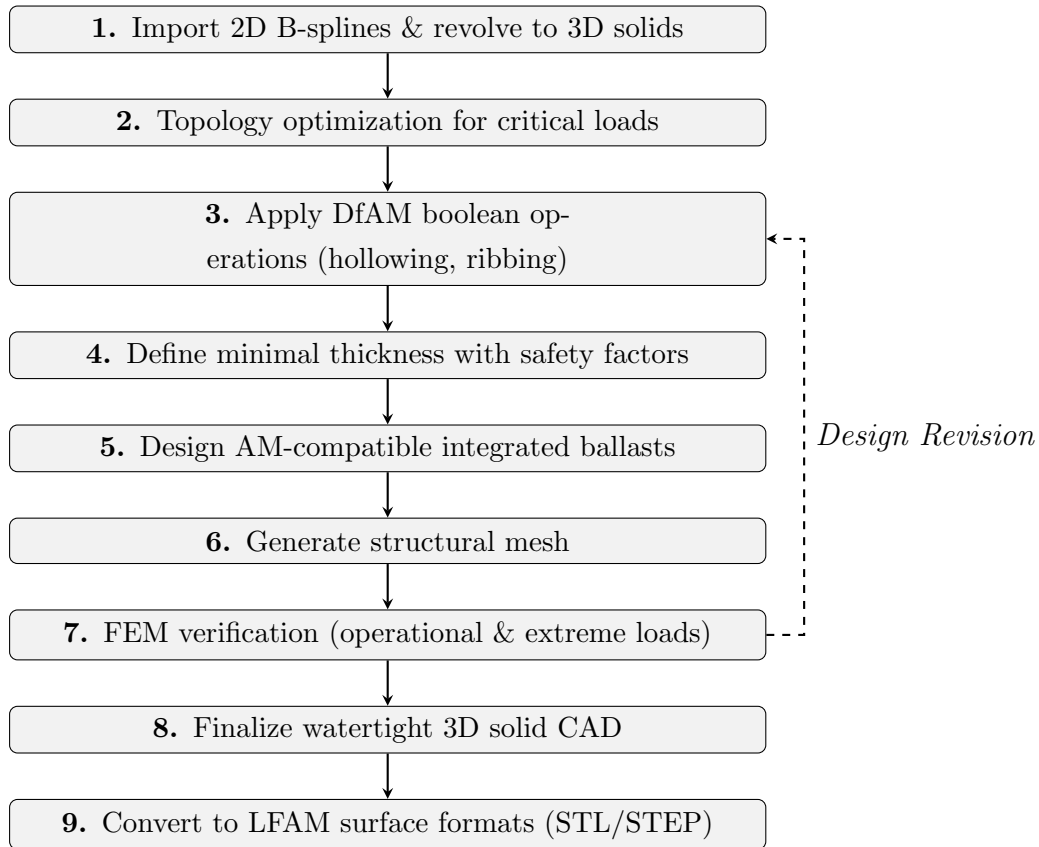


Figure 6: Sequential workflow of the conceptual LFAM Structural Toolbox protocol.

4 Structural Design of the Point Absorber for LFAM

4.1 Hydrodynamic Design and Geometry Initialization

The structural design phase commenced with the importation of the optimal PA geometry (spline 282) from WP1. This 2D profile was imported into FreeCAD and revolved to create the base 3D hydrodynamic envelope (see WP1 report for more details).

4.2 Internal Structural Engineering and Ballast Chambers

A WEC floater is not merely an empty shell; it requires complex internal engineering to function correctly during wave tank testing. As illustrated in Figure 7, the internal volume was systematically segmented to include specific functional features.

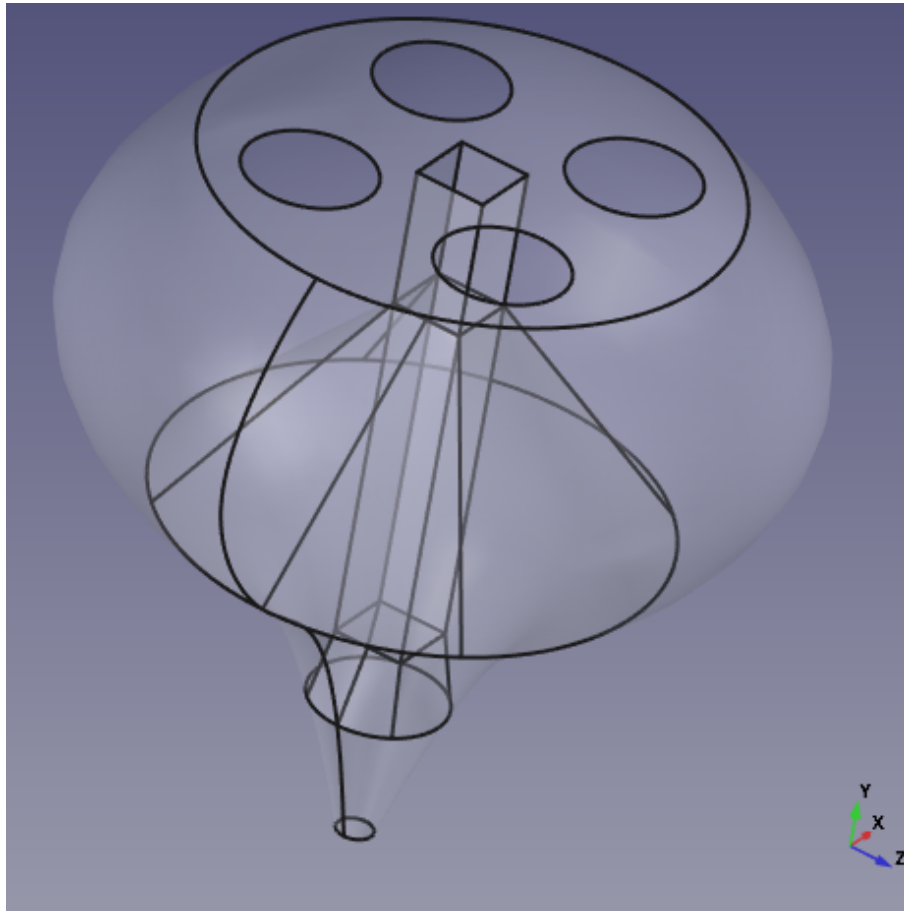


Figure 7: Cross-sectional view of the FreeCAD structural model, detailing the outer hull, the central heave-constraining slot, internal ballast chambers, and top deck sensor access ports.

The internal engineering includes:

- **Central Heave Slot:** A vertical slot was designed directly through the geometric centre of the floater. Crucially, this slot features a *squared cross-sectional area* to allocate a matching vertical bar. This specific geometry physically constrains the PA model strictly to the heave (vertical) degree of freedom during WP3 testing, preventing unwanted rotation or surge.
- **Sensor Access Ports:** Dedicated routing holes and access hatches were integrated into the top deck (cover component). These are essential for installing sensors, allocating data/logging processing devices internally, and allowing for the waterproof routing of telemetry cables.
- **Internal Ballast Chambers:** Two circular ballast chambers were integrated into the design with a dual scope: first, to hold physical ballast (e.g., water or metal weights) to obtain the precise buoy mass and draft required by the

Froude-scaled hydrodynamic model; and second, to act as internal structural ribs, significantly reinforcing the outer hull against wave slamming.

4.3 Design for Additive Manufacturing (DfAM) Constraints

The most critical aspect of the WP2 toolbox is adapting the design for tool-less LFAM. The internal ballast chambers and the main hull were subjected to strict DfAM rules:

1. **Overhang Limits:** LFAM cannot print in mid-air. All internal roofs of the ballast chambers and the transitions to the central slot were designed with self-supporting angles (strictly kept under a $45^\circ - 60^\circ$ overhang relative to the vertical Z-axis) to completely eliminate the need for internal sacrificial support structures.
2. **Upside-Down Print Orientation:** It was determined that the optimal manufacturing strategy is to print the PA floater *upside down*. By placing the flat top deck directly on the print bed, the complex, curved lower hull naturally builds upwards and inwards towards the keel, inherently respecting overhang limits and ensuring a pristine hydrodynamic surface finish on the water-facing exterior.

5 Preliminary Finite Element Method (FEM) Analysis

To ensure the structural integrity of the floater, a preliminary FEM analysis was configured within the FreeCAD toolbox.

A critical output of this analysis was the determination of the global wall thickness. Initial simulations indicated that a minimum thickness of 6 mm was sufficient for the lower hull to withstand baseline hydrostatic pressures, though the flat top deck required slightly thicker dimensions to prevent deflection. However, to ensure robust structural safety—applying a safety factor considering maximum wave loads of 1.8—and to adhere strictly to the specific LFAM extrusion constraints indicated by Caracol’s manufacturing guidelines, the global PA thickness was conservatively set to **10 mm**.

5.1 Material Specifications: PP GF30 (FGF)

The selected material for the LFAM process is Polypropylene reinforced with 30% Glass Fibre (PP GF30) via Fused Granular Fabrication (FGF). This specific material was chosen based on Caracol's previous successful manufacturing experience, notably their successful printing of a functional marine boat (the "Beluga" project). PP GF30 is exceptionally suitable for the marine environment due to its high chemical resistance (preventing saltwater degradation and corrosion) and extremely low moisture absorption. Furthermore, the 30% glass fibre reinforcement provides the necessary structural stiffness to withstand wave impacts while significantly reducing the coefficient of thermal expansion, thereby mitigating thermal warping during the layer-by-layer LFAM process.

Because LFAM inherently creates anisotropic structures (stronger along the extruded bead, weaker across the Z-axis layers), two material configurations were defined for the analysis to represent the upper (longitudinal) and lower (transverse) bounds of structural performance. The properties implemented in the FreeCAD FEM workbench are detailed in Table 1.

Table 1: Material Configuration Table: PP GF30 (FGF)

Panel Section	Property Unit (FreeCAD)		Config 1	Config 2
Basic Properties	Density	kg/m ³	1080	1080
Mechanical Properties	Young's modulus	MPa	5000	2000
	Poisson ratio	-	0.35	0.35
Thermal Properties	Thermal conductivity	W/m/K	0.32	0.32
	Expansion coefficient	$\mu\text{m}/\text{m}/\text{K}$	40	40
	Reference temperature	K	293.15 (20°C)	293.15 (20°C)
	Specific heat capacity	J/kg/K	1550	1550
Failure Limits	Ultimate Tensile Strength	MPa	55 – 70	15 – 25

5.2 Definition of Load Cases

To rigorously validate the Point Absorber design, five specific load cases (LC) were established, simulating both hydrostatic pressure and dynamic wave forces acting on the structure. The load magnitudes were defined based on the expected operational forces: 30 kPa for hydrostatic pressure, 15 kN for lateral loads, and 21 kN for vertical heave loads. The load cases are summarised in Table 2.

Table 2: Definition of Operational Load Cases for FEM Verification

Load Case	Description	Magnitude	Notes
LC1	Hydrostatic Pressure	30 kPa	The deck is considered full/covered, acting as a single watertight component.
LC2	Load [1, 0, 0]	15 kN	Surge/lateral force (X-direction).
LC3	Load [1, 1, 0]	15 kN	Diagonal planar force (XY-direction, 45 degrees).
LC4	Load [0, 0, 1]	21 kN	Positive heave force (bottom load).
LC5	Load [0, 0, -1]	21 kN	Negative heave force (top load).

5.3 Preliminary Results and Verification

The FEM simulations were executed using the CalculiX solver within FreeCAD. For each of the five load cases, the Von Mises stress was extracted to evaluate the structural response of the 10 mm hull and internal features. The goal of this analysis is to verify if the peak stresses remain below the conservative Ultimate Tensile Strength (UTS) limit of **15 MPa**, as defined in Configuration 2 (representing the weaker Z-axis inter-layer adhesion limit inherent to LFAM). The results for each load case are presented below, visualising the stress distributions from various perspectives.

LC1: Hydrostatic Pressure (30 kPa)

Figures 8 and 9 illustrate the Von Mises stress distribution under a uniform 30 kPa hydrostatic pressure. The figures provide both an overall external view of the floater and an internal view detailing the stress distribution across the inner walls and ballast chambers.

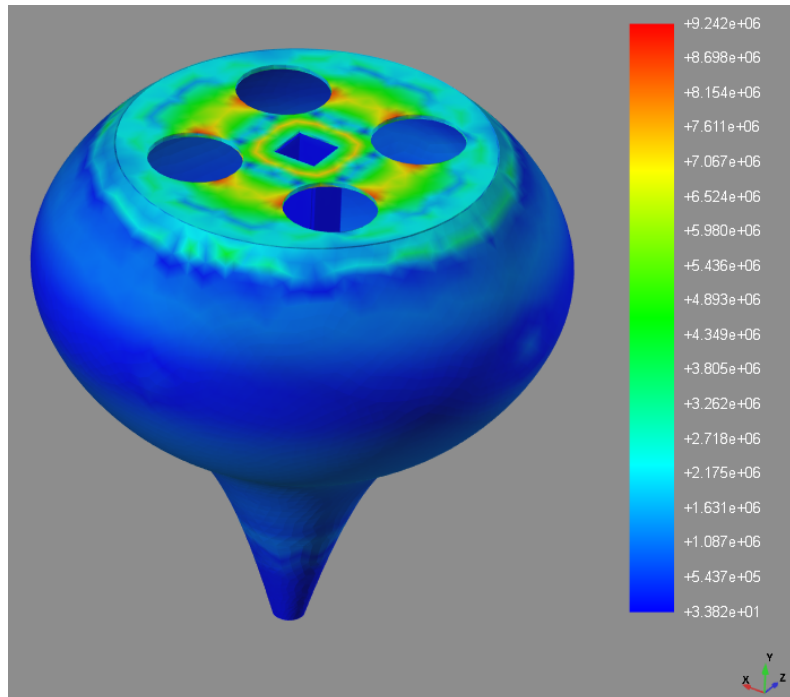


Figure 8: External view of the Von Mises stress distribution under 30 kPa hydrostatic pressure (LC1).

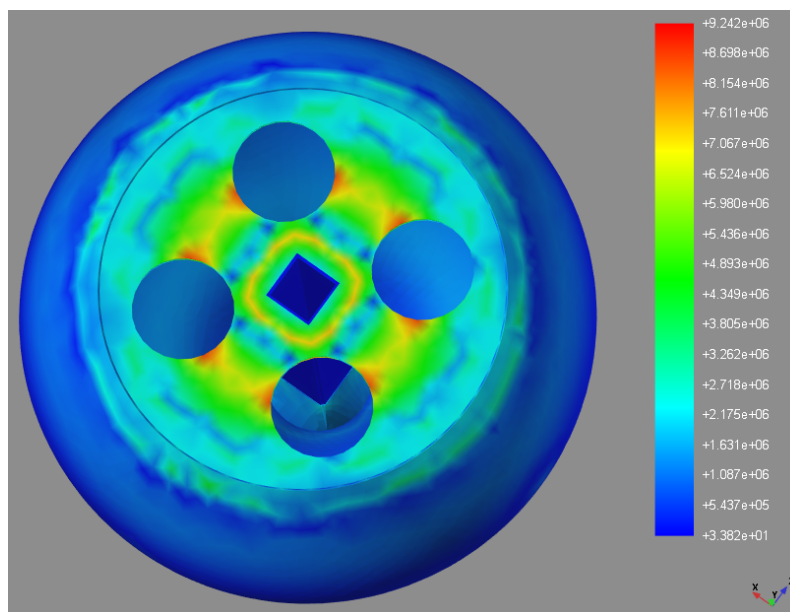


Figure 9: Internal view of the Von Mises stress distribution under 30 kPa hydrostatic pressure (LC1).

LC2: Lateral Surge Force (15 kN)

Figures 10 and 11 show the structural response to a 15 kN lateral load (surge force). The side view highlights the stress on the outer hull profile, while the internal

view demonstrates how the stress propagates through the internal geometry and the central slot.

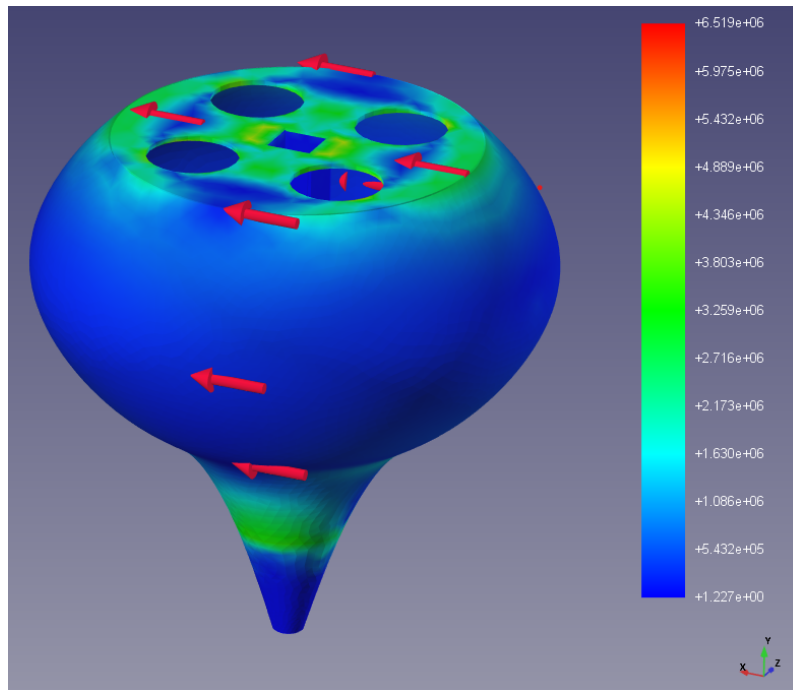


Figure 10: Side view of the Von Mises stress distribution under a 15 kN lateral load (LC2).

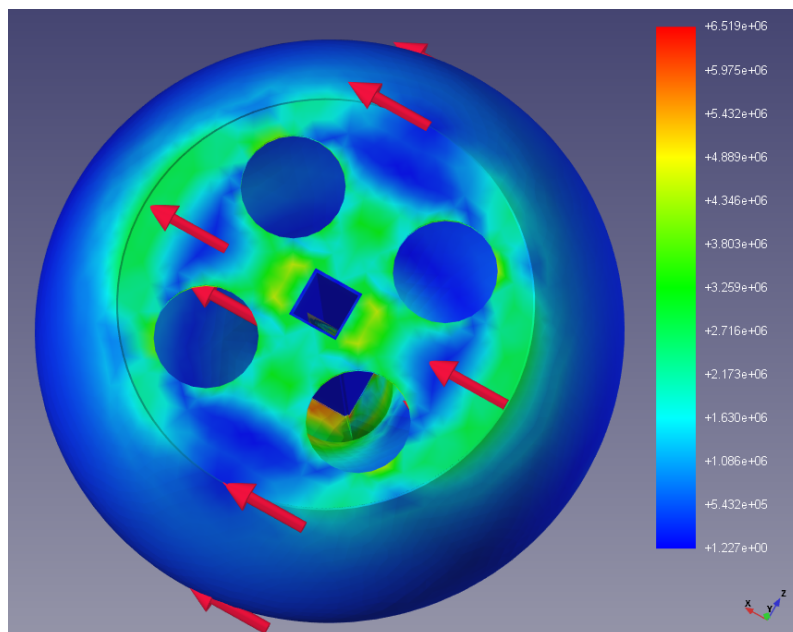


Figure 11: Internal view of the Von Mises stress distribution under a 15 kN lateral load (LC2).

LC3: Diagonal Lateral Force (15 kN at 45 degrees)

Figure 12 illustrates the internal view of the Von Mises stress when the 15 kN lateral load is applied at a 45-degree angle, testing the asymmetric response of the internal chambers.

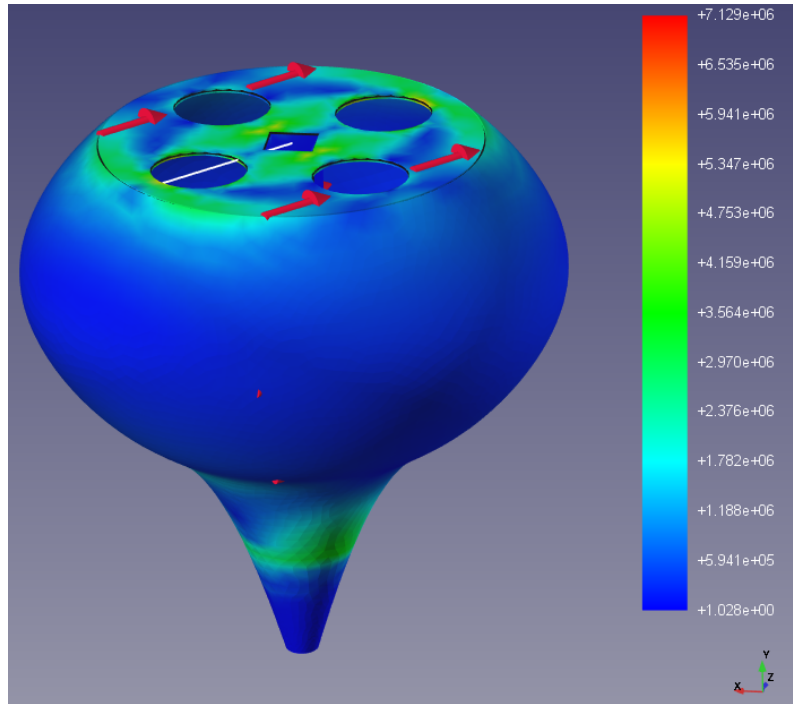


Figure 12: Internal view of the Von Mises stress distribution under a 15 kN lateral load applied at 45 degrees (LC3).

LC4: Positive Heave / Bottom Load (21 kN)

Figures 13, 14, and 15 present the results for a 21 kN load applied from the bottom (simulating positive heave). The figures detail the Von Mises stress from the bottom perspective, a lateral perspective, and an internal cross-section.

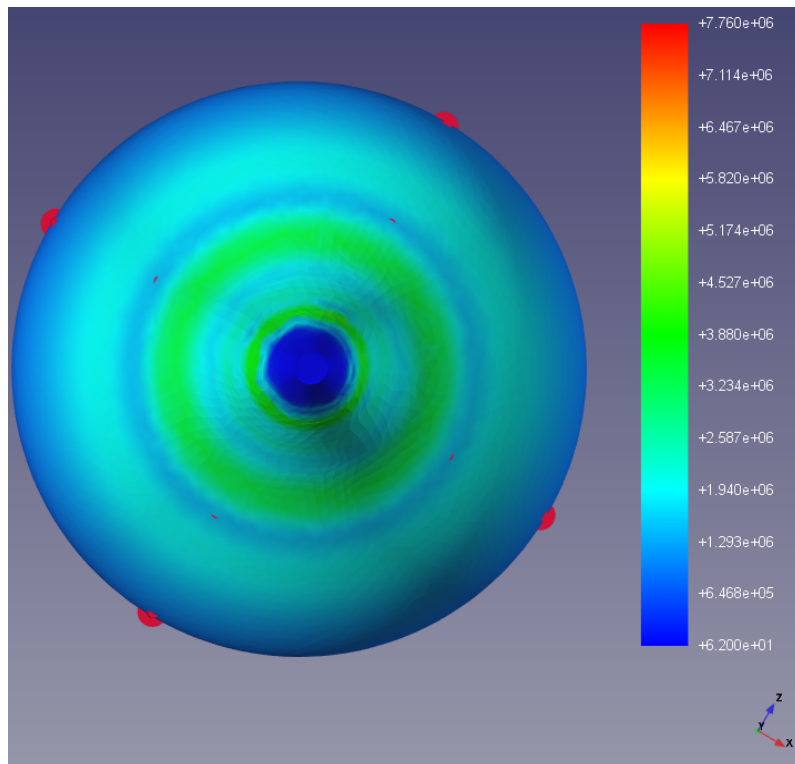


Figure 13: Bottom view of the Von Mises stress distribution under a 21 kN bottom load (LC4).

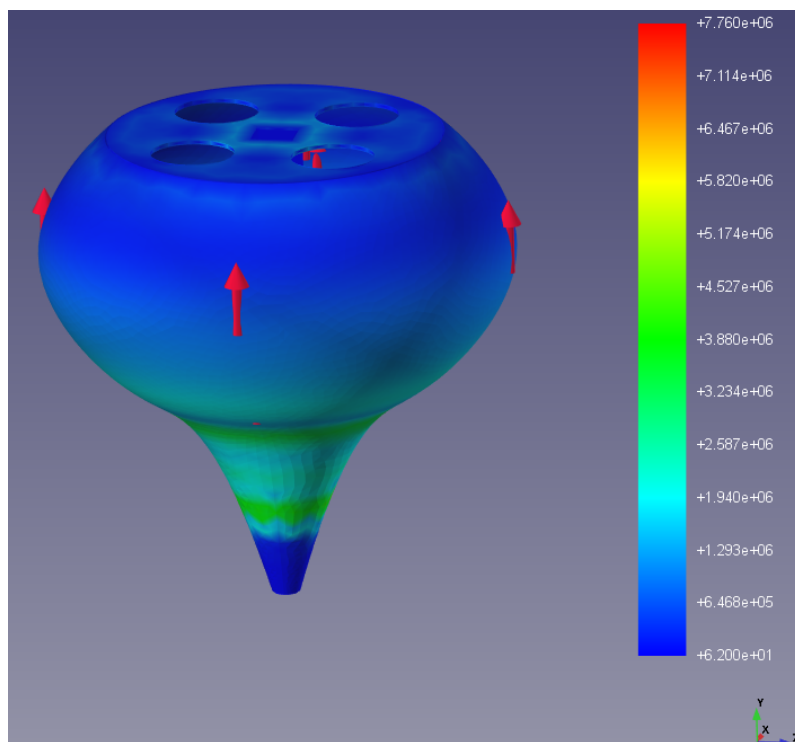


Figure 14: Lateral view of the Von Mises stress distribution under a 21 kN bottom load (LC4).

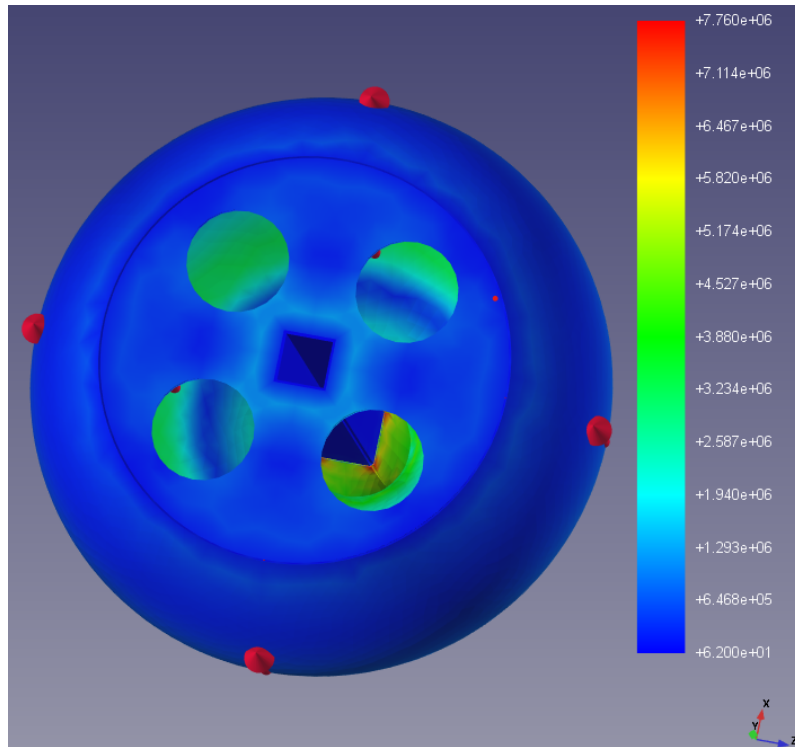


Figure 15: Internal view of the Von Mises stress distribution under a 21 kN bottom load (LC4).

LC5: Negative Heave / Top Load (21 kN)

Finally, Figures 16 and 17 display the Von Mises stress distribution resulting from a 21 kN load applied from the top (simulating negative heave). The results are shown from both a bottom view and an internal view to assess the load transfer through the top deck and central column.

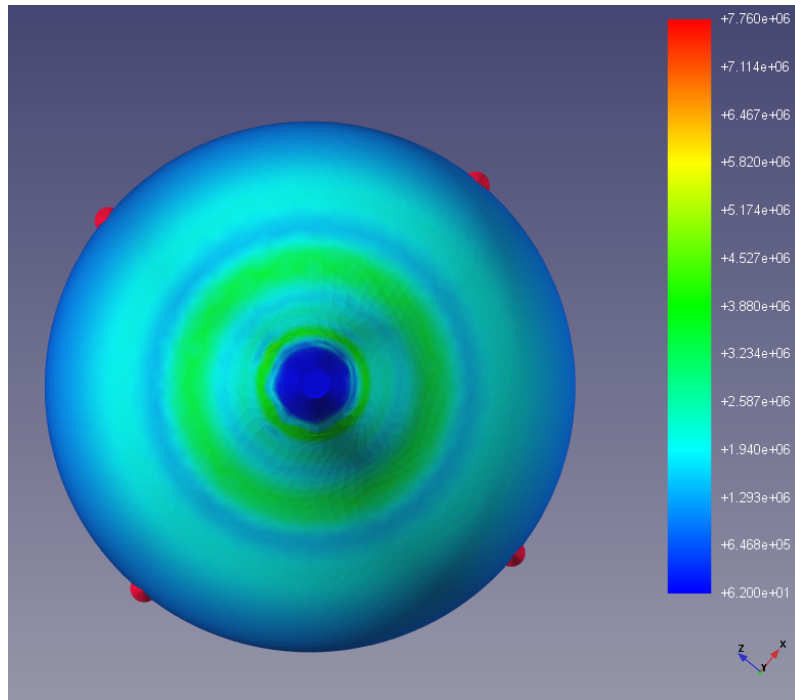


Figure 16: Bottom view of the Von Mises stress distribution under a 21 kN top load (LC5).

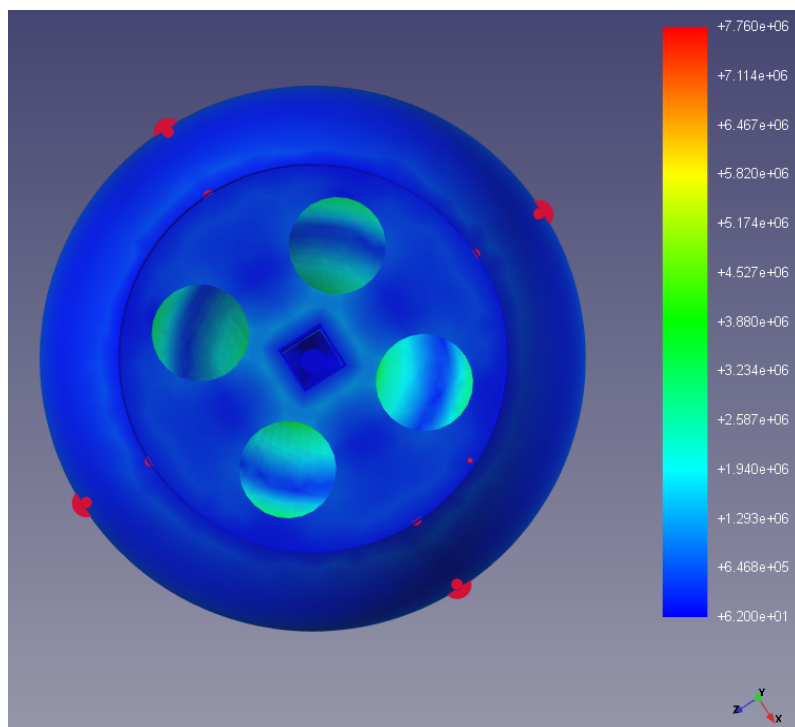


Figure 17: Internal view of the Von Mises stress distribution under a 21 kN top load (LC5).

5.4 PA and Modelling Specifications

To provide a comprehensive overview of the physical and hydrodynamic parameters driving the structural design, the full specifications of the Point Absorber (derived from the optimal spline 282 geometry and the PTO sizing runs) are summarised in Table 3.

Table 3: Full Specifications of the Optimised Point Absorber (PA)

Parameter	Description	Value
Diameter	Outer diameter of the floater at the waterline	1.0 m
Draft	Submerged depth of the buoy	0.75 m
Deck Height	Freeboard/height of the deck above waterline	0.25 m
Displacement	Total displaced mass of the floater	198.5 kg
Global Max Force	Worst-case 3-hour peak force ($F_{max.3hr}$)	10.63 kN
RMS Force	Root Mean Square force (F_{rms})	2.77 kN
Max Wave Height	Significant wave height (H_s) for worst-case	2.125 m
JONSWAP γ	Peak enhancement factor for wave spectrum	3.3
Hydrostatic Stiffness	Restoring force coefficient (K_{hydro})	7898.2 N/m
BEM Solver	Hydrodynamic solver version used	NEMOH v3.0.2
Mesh Panels	Number of panels used in BEM evaluation	250

6 Conclusion and Integration with WP3

Work Package 2 successfully established a pragmatic, FreeCAD-based structural toolbox bridging the gap between hydrodynamic optimisation and Large-Format Additive Manufacturing. By transitioning to the Point Absorber archetype, the project leveraged mature operational data to design a highly functional, test-ready

physical model. The strict adherence to DfAM constraints—specifically the integration of self-supporting internal ballast chambers and the strategic upside-down print orientation—ensures that the PA floater can be manufactured without wasteful support structures. With the preliminary FEM verification executed to evaluate the integrity of the 10 mm PP GF30 hull across all critical load cases, the digital design is fully prepared for physical extrusion and subsequent hydrodynamic wave tank testing, which is detailed in the forthcoming WP3 report.